

IBADULLAH 25K-6544

QUESTION 01:

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int arr[5];
```

```
    int i, temp;
```

```
    printf("Enter 5 integers:\n");
```

```
    for(i = 0; i < 5; i++)
```

```
    {
```

```
        scanf("%d", &arr[i]);
```

```
    }
```

```
    temp = arr[4];
```

```
    for(i = 4; i > 0; i--)
```

```
    {
```

```
        arr[i] = arr[i-1];
```

```
    }
```

```
    arr[0] = temp;
```

```
    printf("Output: ");
```

```
    for(i = 0; i < 5; i++)
```

```
    {
```

```
        printf("%d ", arr[i]);
```

```
    }
```

```
    return 0;
```

```
}
```

```

Enter 5 integers:
3
4
1 5
1 6
1 7
1 Output: 7 3 4 5 6
1 -----
1 Process exited after 5.306 seconds
1 Press any key to continue . . . |
1

```

QUESTION 02:

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int arr[10];
```

```
    int i, num, count = 0;
```

```
    printf("Enter 10 numbers:\n");
```

```
    for(i = 0; i < 10; i++)
```

```
    {
```

```
        scanf("%d", &arr[i]);
```

```
    }
```

```
    printf("Enter number to search: ");
```

```
    scanf("%d", &num);
```

```
    for(i = 0; i < 10; i++)
```

```
    {
```

```
        if(arr[i] == num)
```

```
        {
```

```
            count++;
```

```
        }
```

```
    }
```

```
    if(count == 0)
```

```
        printf("number not found");
```

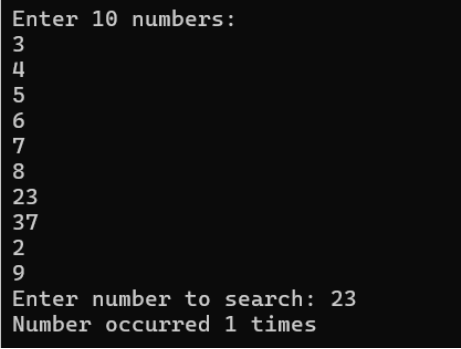
```

else

    printf("Number occurred %d times", count);

return 0;
}

```



```

Enter 10 numbers:
3
4
5
6
7
8
23
37
2
9
Enter number to search: 23
Number occurred 1 times

```

QUESTION 03:

```

#include <stdio.h>

int main()
{
    int pass[10], fail[10];
    int i = 0, j = 0;
    int mark;
    float avgPass = 0, avgFail = 0;
    printf("Enter marks (-1 to exit):\n");
    while(1)
    {
        scanf("%d", &mark);
        if(mark == -1)
            break;

        if(mark >= 5 && mark <= 10 && i < 10)
        {
            pass[i] = mark;
            avgPass += mark;

```

```

        i++;
    }
    else if(mark >= 0 && mark < 5 && j < 10)
    {
        fail[j] = mark;
        avgFail += mark;
        j++;
    }
}

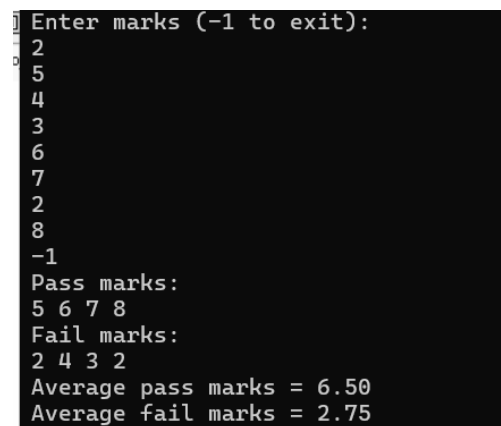
printf("Pass marks:\n");
for(int k = 0; k < i; k++)
    printf("%d ", pass[k]);
printf("\nFail marks:\n");
for(int k = 0; k < j; k++)
    printf("%d ", fail[k]);

if(i > 0)
    printf("\nAverage pass marks = %.2f", avgPass/i);

if(j > 0)
    printf("\nAverage fail marks = %.2f", avgFail/j);

return 0;
}

```



```

Enter marks (-1 to exit):
2
5
4
3
6
7
2
8
-1
Pass marks:
5 6 7 8
Fail marks:
2 4 3 2
Average pass marks = 6.50
Average fail marks = 2.75

```

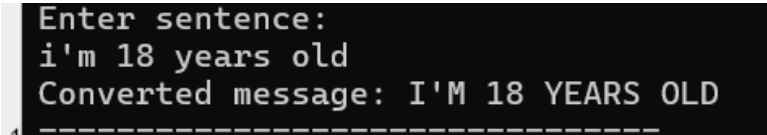
QUESTION 04:

```

#include <stdio.h>

int main()
{
    char str[100];
    int i = 0;
    printf("Enter sentence:\n");
    scanf("%[^\\n]", str);
    while(str[i] != '\\0')
    {
        if(str[i] >= 'a' && str[i] <= 'z')
            str[i] = str[i] - 32;
        else if(str[i] >= 'A' && str[i] <= 'Z')
            str[i] = str[i] + 32;
        i++;
    }
    printf("Converted message: %s", str);
    return 0;
}

```



```

Enter sentence:
i'm 18 years old
Converted message: I'M 18 YEARS OLD

```

QUESTION 05:

```

#include <stdio.h>

int main()
{
    int arr[10];
    int i, max, min;
    printf("Enter 10 integers:\n");
    for(i = 0; i < 10; i++)
    {

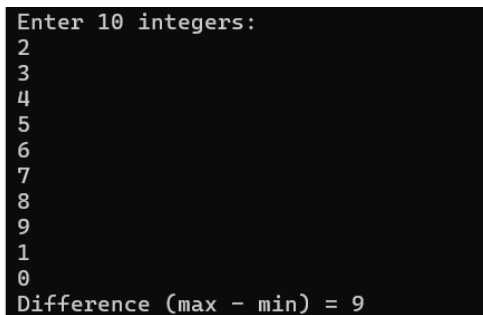
```

```

        scanf("%d", &arr[i]);
    }
    max = min = arr[0];
    for(i = 1; i < 10; i++)
    {
        if(arr[i] > max)
            max = arr[i];

        if(arr[i] < min)
            min = arr[i];
    }
    printf("Difference (max - min) = %d", max - min);
    return 0;
}

```



```

Enter 10 integers:
2
3
4
5
6
7
8
9
1
0
Difference (max - min) = 9

```

QUESTION 06:

```

#include <stdio.h>

int main()
{
    char str[100];
    int i = 0;
    int vowels = 0, consonants = 0;
    printf("Enter a word:\n");
    scanf("%s", str);
    while(str[i] != '\0')

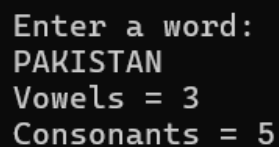
```

```

{
    if(str[i]=='a' || str[i]=='e' || str[i]=='i' || str[i]=='o' || str[i]=='u' ||
        str[i]=='A' || str[i]=='E' || str[i]=='I' || str[i]=='O' || str[i]=='U')
    {
        vowels++;
    }
    else
    {
        consonants++;
    }
    i++;
}

printf("Vowels = %d\n", vowels);
printf("Consonants = %d\n", consonants);
return 0;
}

```



```

Enter a word:
PAKISTAN
Vowels = 3
Consonants = 5

```

QUESTION 07:

```

#include <stdio.h>

int main()
{
    int arr[10];
    int i, j;
    printf("Enter 10 integers:\n");
    for(i = 0; i < 10; i++)
    {
        scanf("%d", &arr[i]);
    }
}

```

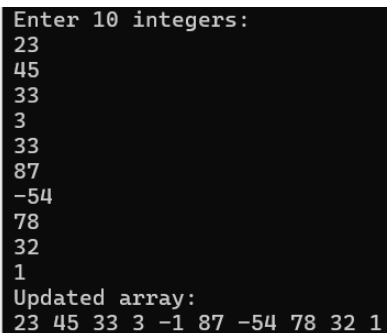
```

        for(j = 0; j < i; j++)
        {
            if(arr[i] == arr[j])
            {
                arr[i] = -1;
                break;
            }
        }
    }

    printf("Updated array:\n");
    for(i = 0; i < 10; i++)
    {
        printf("%d ", arr[i]);
    }

    return 0;
}

```



```

Enter 10 integers:
23
45
33
3
33
87
-54
78
32
1
Updated array:
23 45 33 3 -1 87 -54 78 32 1

```

QUESTION 08:

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    char str[100];
```

```
    printf("Enter characters (non alphabets only will be stored):\n");
```

```
    scanf("%[^A-Za-z]", str);
```



```
printf("Non alphabet characters entered:\n%s", str);  
return 0;  
}
```